



Inspiring Excellence

CSE370: Database Systems Project Report

Project Title: Disaster Relief Management System

Group No: 8, CSE370 Lab Section: 11, Summer 2026		
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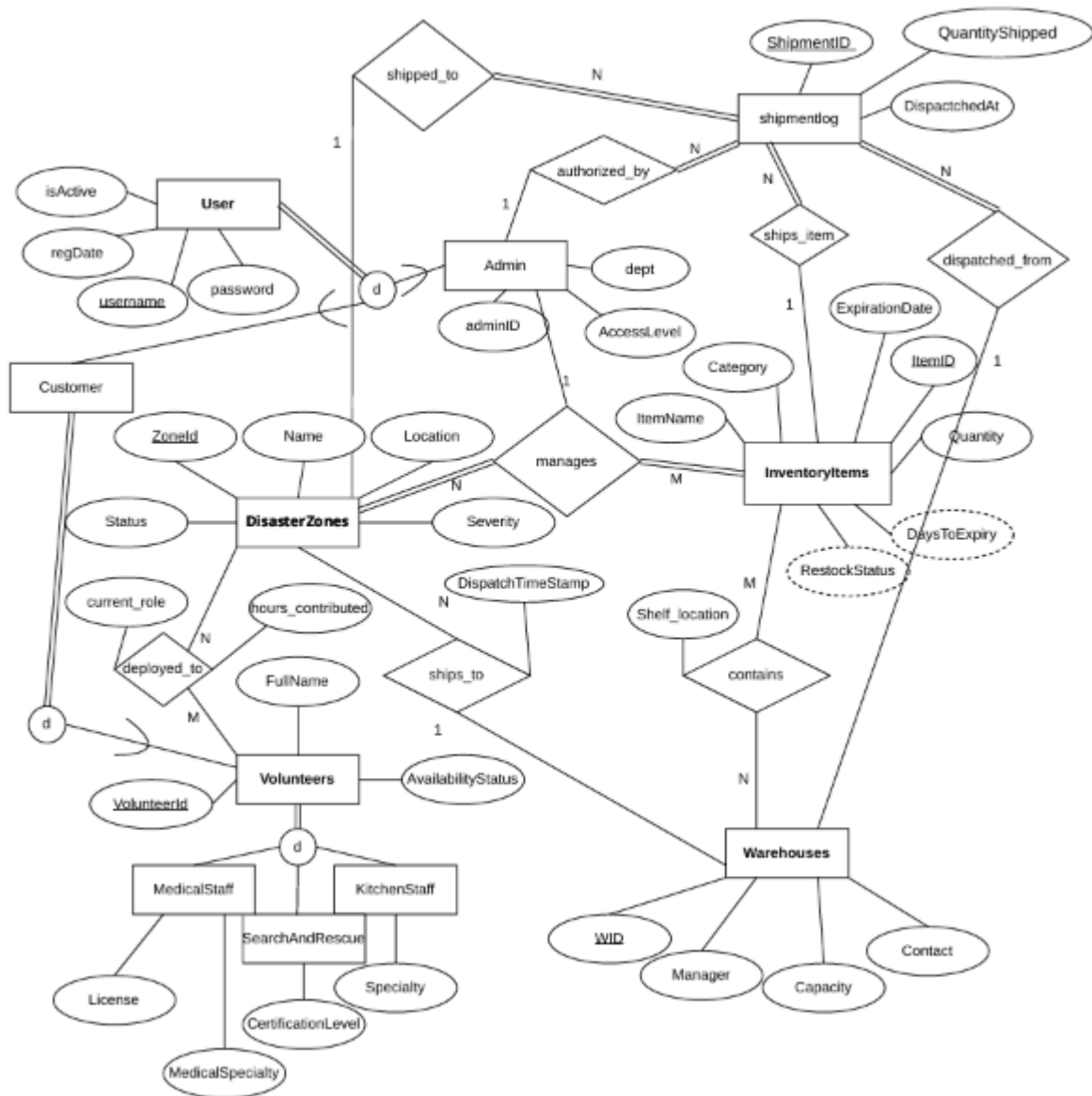
Introduction

The idea behind our DBMS project is “Disaster Relief System” which takes a real life problem of civilians suffering from natural disasters and creates a prototype of a solution that can manage the relief system during these times. Our system will allow users to view disaster zones and sign up as volunteers to help victims.

Project Features

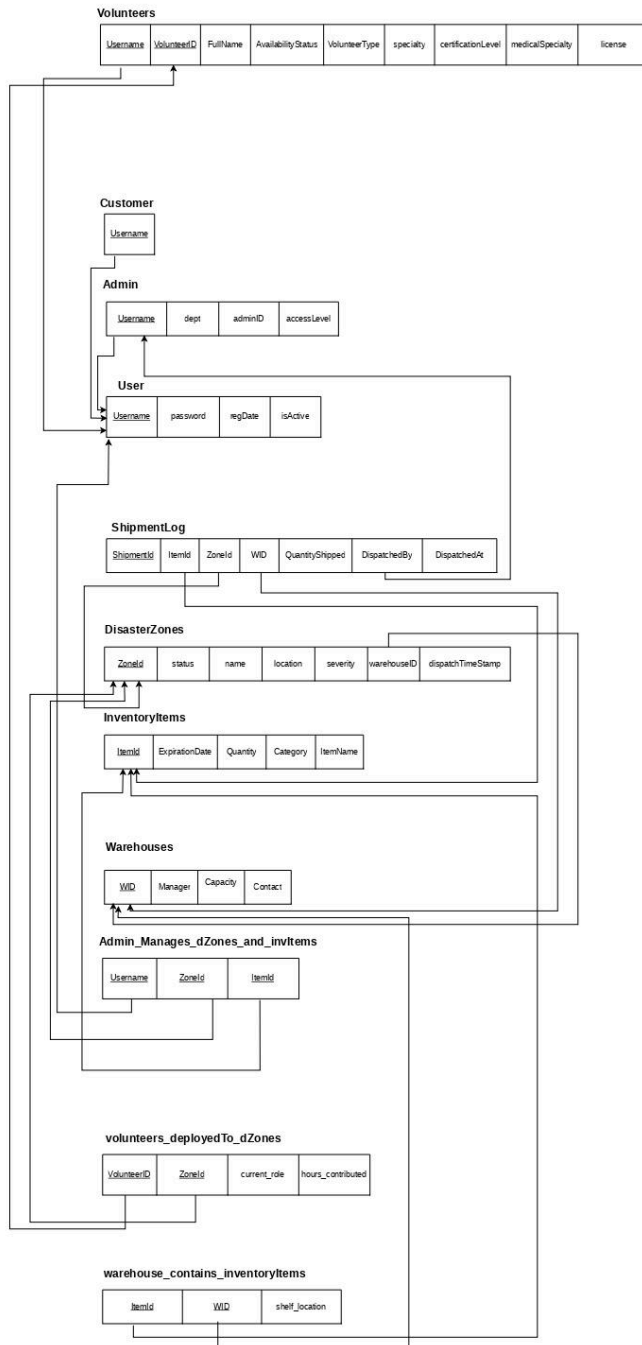
ID, Name	Features	
	Ft 1	Login
24301374, Shadeed Aarshan	Ft 2	Customers can view disaster zones and apply to be volunteers
	Ft 3	Admin can approve/delete volunteers
	Ft 4	Admin can deploy/release volunteers to and from disaster zones
	Ft 5	Volunteers can toggle availability, controlling whether they can be deployed or not
24301348, Kazi Aryan Rahman	Ft 1	Admin can declare, view, and update disaster zones, assigning a severity level and source warehouse
	Ft 2	Admin can dispatch relief supplies from a warehouse to a disaster zone, with automatic stock deduction and shipment logging
	Ft 3	Admin can view live per-zone analytics — active personnel, accumulated hours, and supplies dispatched
23101997, DEF	Ft 1	Warehouse Management and Inventory Assignment: Register warehouses, view/remove/edit warehouses and shelf locations
	Ft 2	Inventory Stock Monitoring and Low Stock Management: view inventory items, add/remove/update inventory
	Ft 3	Expiry Date Auditor: view remaining days until expiry, flag expired, and soon-to-be expired items

ER/EER Diagram



Schema Diagram

SCHEMA DIAGRAM FOR DISASTER MANAGEMENT RELIEF SYSTEM (1NF)



Normalization

- a. Explain if your converted Schema is in 1NF or not. If not, decompose it to 1NF.
- b. Explain if your converted Schema is in 2NF or not. If not, decompose it to 2NF. Can there be any partial functional dependencies in your relational schema?
- c. Explain if your converted Schema is in 3NF or not. If not, decompose it to 3NF. Can there be any transitive dependencies in your relational schema?

Answer:

- a. During schema mapping, it is automatically converted to 1NF as there are no multivalued or composite attributes
- b. Yes, it is in 2NF because there are no functional dependencies
- c. Yes, it is in 3NF because there are no transitive dependencies

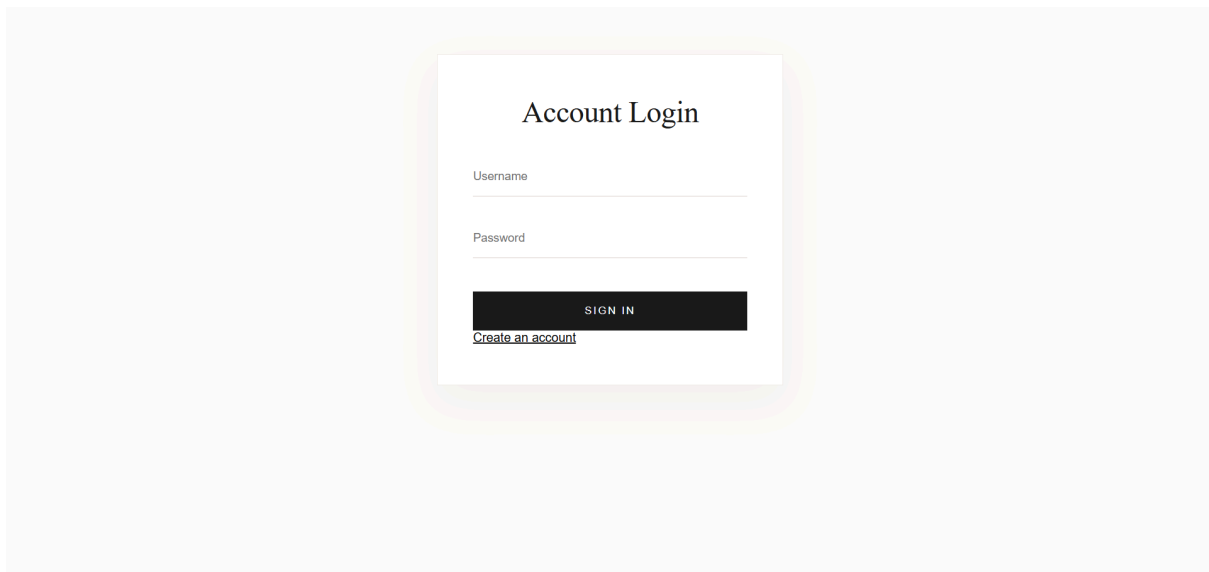
Frontend Development

Briefly discuss about Backend Development and add relevant Screenshots (if required) by mentioning Individual Contributions

Contribution of ID: 24301374, Name: Shadeed Aarshan

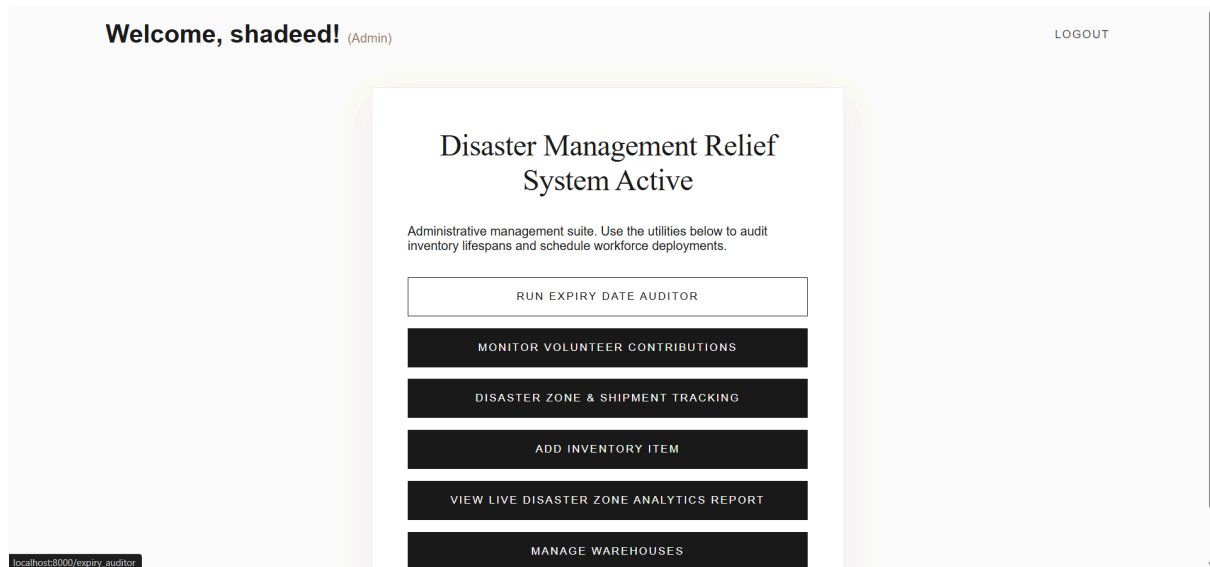
In the backend section, I designed the following pages:

Login page:

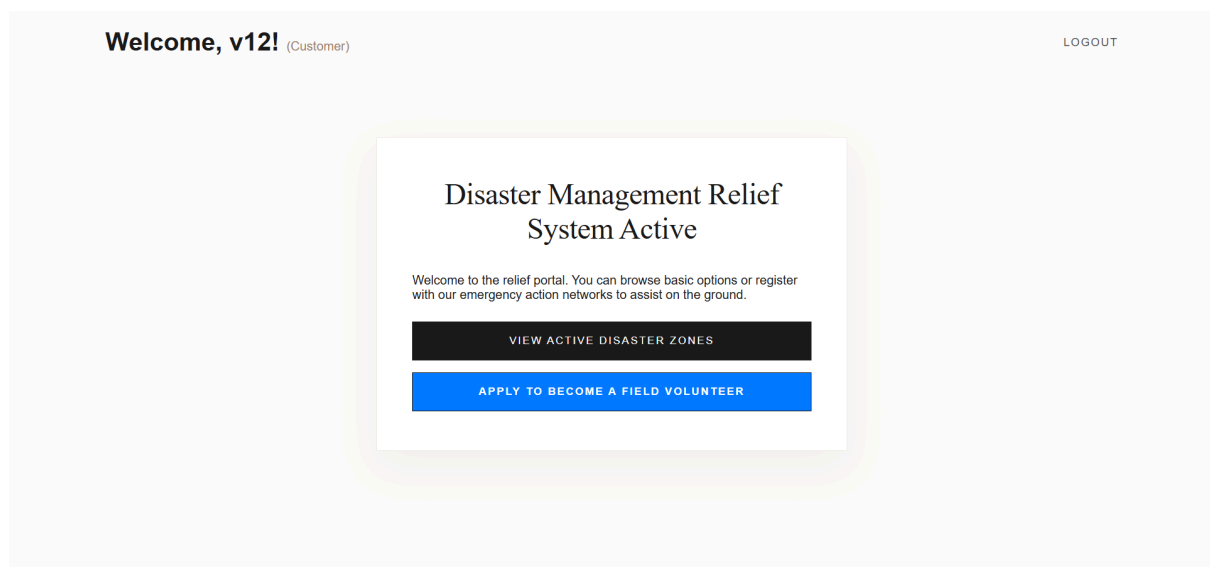


I contributed in designing the admin home page alongside my groupmates

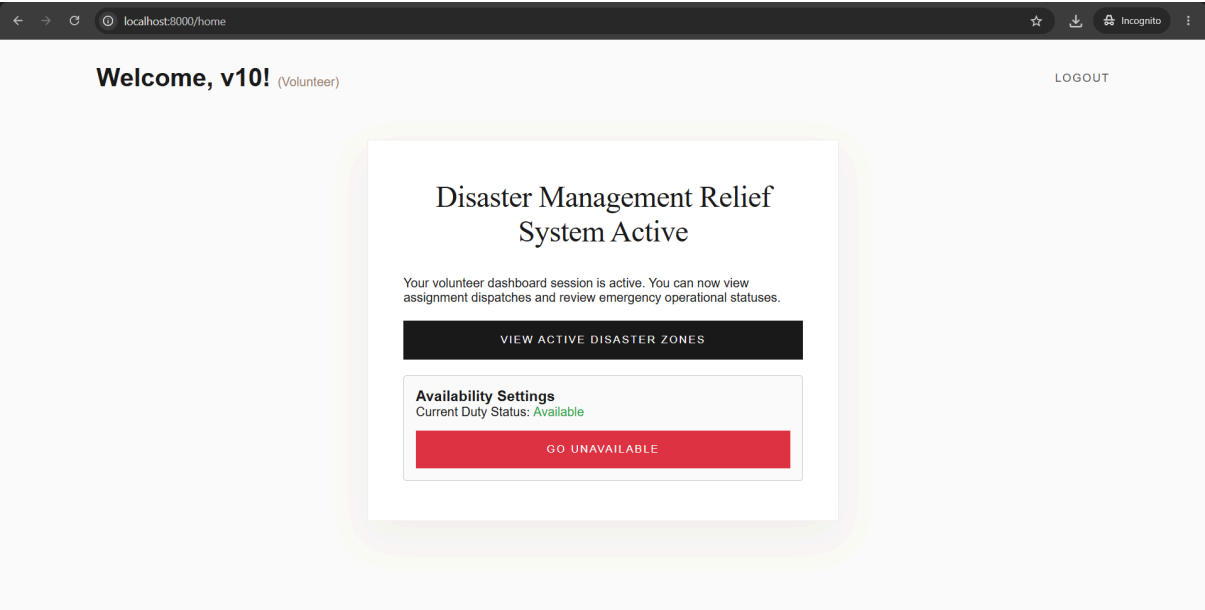
Home page(if admin): admin personnel can navigate their various features



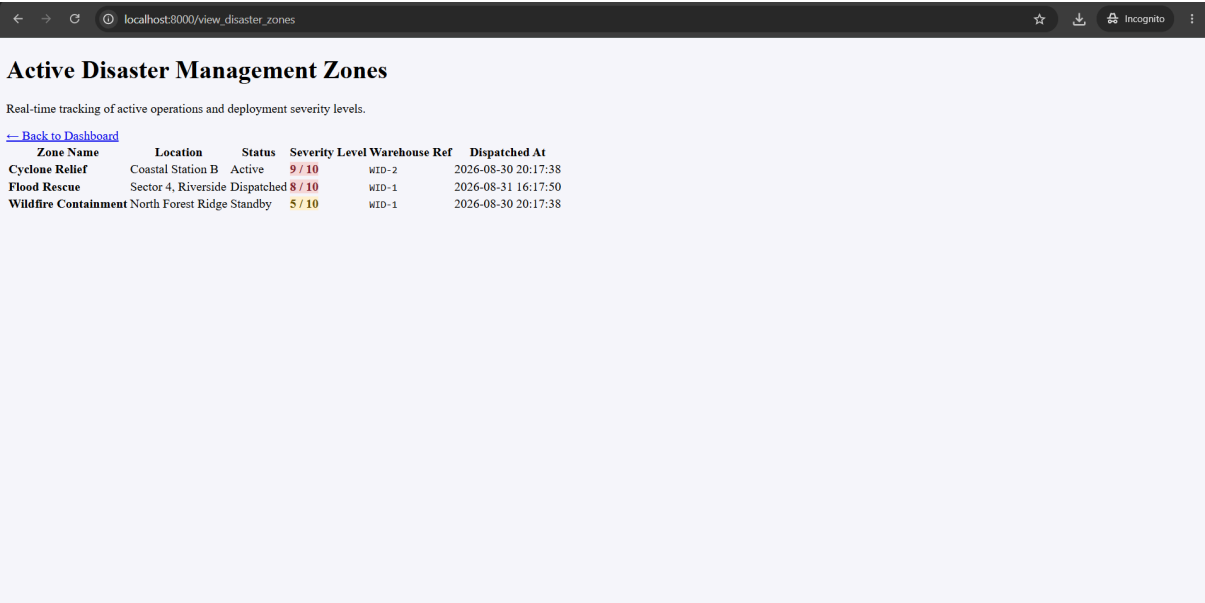
If customer but not volunteer: customers can choose to view active disaster zones or apply to become a volunteer.



If volunteer: Volunteers can toggle their availability status.



View_disaster_zones page:



Volunteer_monitoring: Admin personnel can monitor volunteers, approve/remove them, see who is available and deploy them. They can also view active deployments.

localhost:8000/volunteer_monitoring

Volunteer Monitoring Panel

DASHBOARD ACTIVE DEPLOYMENTS LOGOUT

Personnel & Contribution Ledger

Review application profiles, authorize incoming workforce additions, or execute personnel database adjustments.

ID	Username	Full Name	Specialty	Status	Total Hours	Management Actions
204	v4	Volunteer4	Flood Response	Available	4 hrs	Deploy Remove
208	v8	Volunteer8	Communications	Unavailable	3 hrs	Remove
206	v6	Volunteer6	Mass Care	Available	1 hrs	Deploy Remove
201	v1	Volunteer1	Search & Rescue	Deployed	0 hrs	Remove
207	v7	Volunteer7	Search & Rescue	Deployed	0 hrs	Remove
202	v2	Volunteer2	First Aid	Available	0 hrs	Deploy Remove
203	v3	Volunteer3	Supply Management	Deployed	0 hrs	Remove
209	v9	Volunteer9	Psychological First Aid	Deployed	0 hrs	Remove

Active_deployments: Admin personnel can view current deployments. They can also release the volunteers from deployment.

localhost:8000/active_deployments

Active Field Deployments

VOLUNTEER LEDGER DASHBOARD LOGOUT

Current Mission Workloads

Review live deployments below. Clicking release will automatically calculate runtime hours based on system clock durations.

Volunteer	Disaster Zone	Location	Assigned Role	Deployment Start Time	Operational Status
Volunteer1	Flood Rescue	Sector 4, Riverside	Search & Rescue	2026-08-31 22:44:10	End Mission & Release
Volunteer3	Cyclone Relief	Coastal Station B	Supply Management	2026-08-31 22:44:28	End Mission & Release
Volunteer7	Cyclone Relief	Coastal Station B	Search & Rescue	2026-08-31 22:44:03	End Mission & Release
Volunteer9	Cyclone Relief	Coastal Station B	Emergency Driver	2026-08-31 22:44:19	End Mission & Release

Deploy_volunteer: Here, admin can choose the options and details regarding deploying a volunteer.

← → ↻ localhost:8000/deploy_volunteer?volunteer_id=204 ☆ ⬇️ 🔒 Incognito ⋮

Deployment Dispatch Center

BACK TO MONITORING LOGOUT

Assign Personnel to Field Operations

Select an active emergency destination zone and define the operational duties for the selected field assets.

Selected Field Volunteer

Volunteer4 (Specialty: Flood Response) ▼

Target Disaster Crisis Zone

-- Choose an Active Disaster Zone -- ▼

Assigned Field Operations Role

Primary Specialty: Flood Response ▼

AUTHORIZE & EXECUTE DEPLOYMENT

Become_volunteer: Menu for customers to become a volunteer by providing relevant information.

← → ↻ localhost:8000/become_volunteer ☆ ⬇️ 🔒 Incognito ⋮

Disaster Relief System

DASHBOARD LOGOUT

Volunteer Registry

Full Name

Volunteer Classification

Field Support ▼

General Specialty

Search & Rescue ▼

Certification Level

Level 1 (Entry) ▼

Medical Specialty

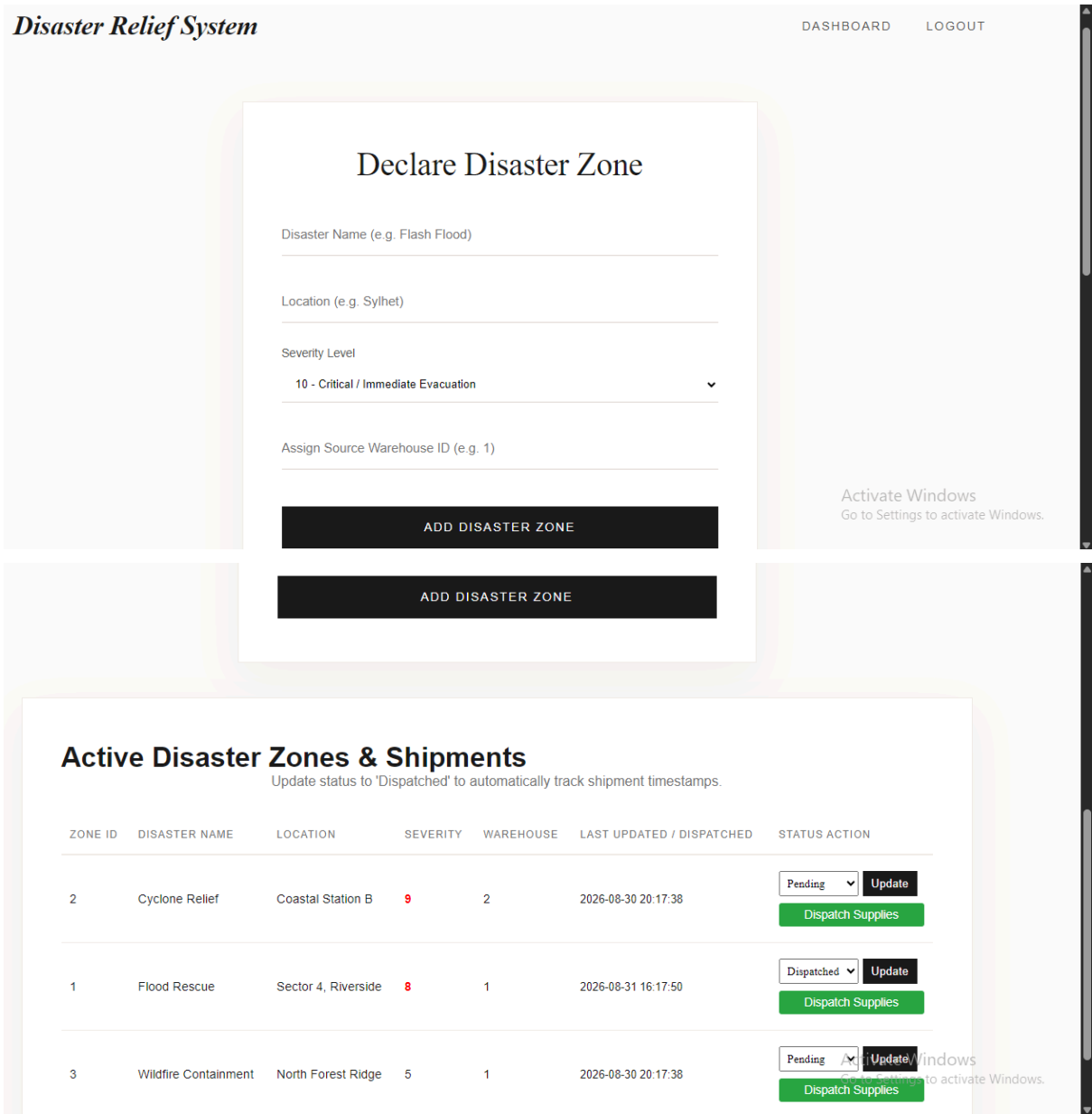
None (Non-Medical) ▼

Professional License Type

Contribution of ID: 24301348, Name: Kazi Aryan Rahman

In the frontend section, I designed the following administrative interfaces:

1. **manage_zones page (Disaster Zone & Shipment Tracking):** Interface for admins to declare new disaster zones and update the operational statuses of active ones.



2. dispatch_shipment Page: Interface to validate and select available inventory from an assigned warehouse to dispatch to a specific crisis zone.

Disaster Relief System

BACK TO ZONESLOGOUT

Dispatch Supplies

Sending supplies to Flood Rescue (Sector 4, Riverside).
Pulling from assigned Warehouse ID: 1

Select Available Item

apples (Available: 5)

Dispatch Quantity

2

AUTHORIZE SHIPMENT

3. zone_analytics Page: Dashboard displaying a real-time, aggregated report on active volunteer deployments and total dispatched supplies per zone.

Disaster Relief System

DASHBOARDLOGOUT

Live Disaster Zone Analytics

Live resource and personnel effort report grouped by active zones.

ZONE ID	CRISIS NAME	LOCATION	SEVERITY	CURRENT STATUS	ACTIVE VOLUNTEERS DEPLOYED	LIVE VOLUNTEER HOURS CONTRIBUTED	TOTAL SUPPLIES DISPATCHED
2	Cyclone Relief	Coastal Station B	LEVEL 9	Active	4 Personnel	12 Hours	0 Items
1	Flood Rescue	Sector 4, Riverside	LEVEL 8	Dispatched	1 Personnel	3 Hours	5 Items
3	Wildfire Containment	North Forest Ridge	LEVEL 5	Standby	0 Personnel	0 Hours	0 Items

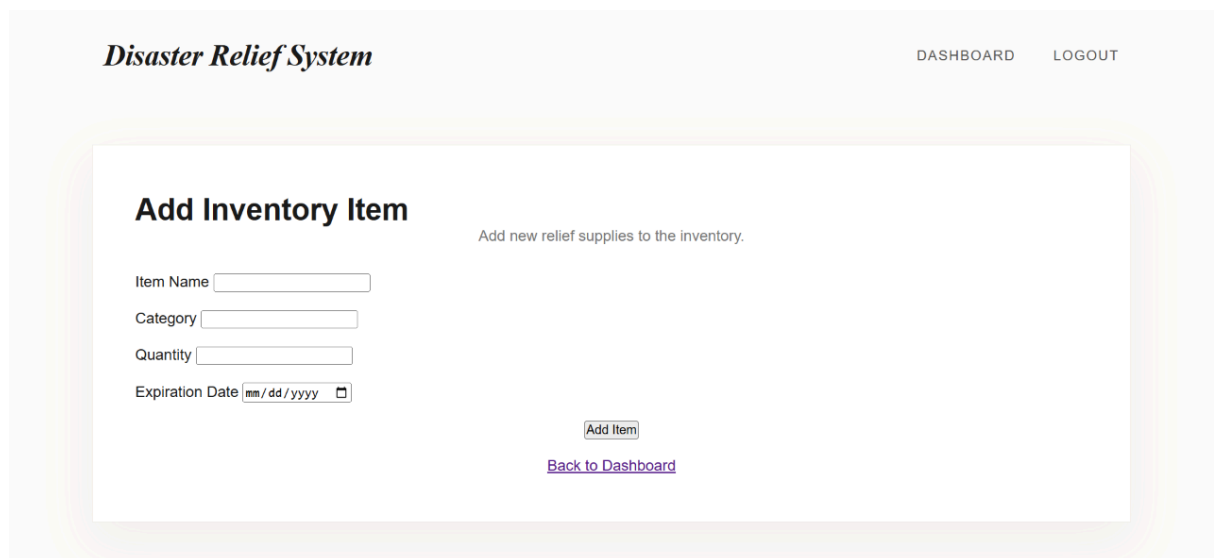
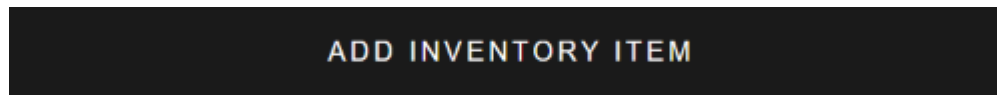
Activate Windows
Go to Settings to activate Windows

Contribution of ID: 24301362, Name: Nuzhat Tabassum Oishee

In the frontend section, I designed the following pages:

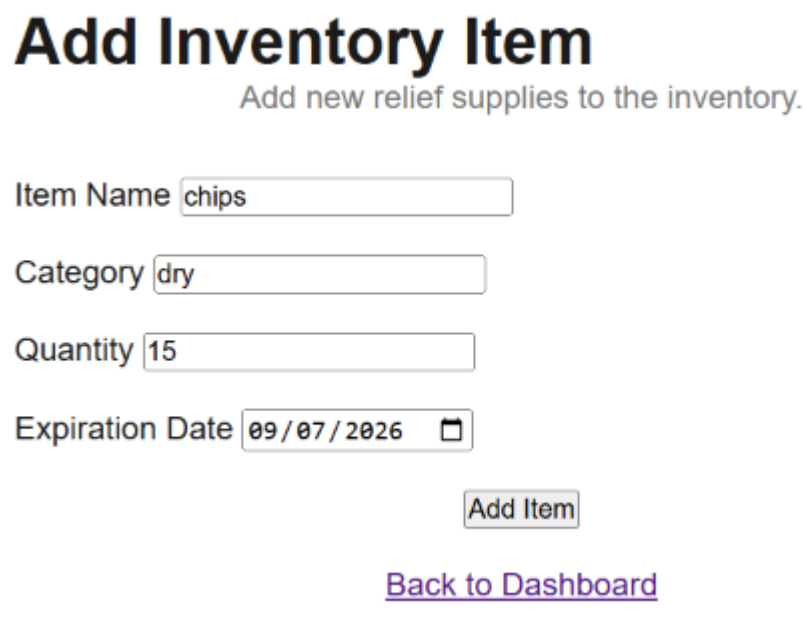
1. Add_inventory page

This page adds an inventory item (which is seen in the expiry date auditor after being added) based on item name, category, quantity, and expiration date



The screenshot shows a web application titled "Disaster Relief System" with a navigation bar containing "DASHBOARD" and "LOGOUT". The main content area is titled "Add Inventory Item" and includes the subtitle "Add new relief supplies to the inventory." Below this, there are four input fields: "Item Name", "Category", "Quantity", and "Expiration Date" (with a date picker icon). To the right of the "Expiration Date" field is a small "Add Item" button. Below the input fields is a purple link labeled "Back to Dashboard".

Adding an item, eg., chips here, will cause it to show up in the expiry date auditor



This image shows a detailed view of the "Add Inventory Item" form. The title "Add Inventory Item" is prominently displayed at the top, followed by the subtitle "Add new relief supplies to the inventory." The form contains four input fields: "Item Name" with the value "chips", "Category" with the value "dry", "Quantity" with the value "15", and "Expiration Date" with the value "09/07/2026" and a calendar icon. Below the input fields is a grey button labeled "Add Item". At the bottom of the form is a purple link labeled "Back to Dashboard".

What we see in the expiry date auditor after adding it:

7	chips	dry	15	STOCK AVAILABLE	2026-09-07	URGENT (6 DAYS LEFT)	Amount	+ Add	- Remove	Edit	Delete
---	-------	-----	----	-----------------	------------	----------------------	--------	-------	----------	------	--------

2. Expiry date auditor page

The goal of this page is to view the added inventory items based on their shelf lives. We can see every inventory item's name, category, quantity, expiration date and days left to expire. Moreover, based on their quantity, it tells us whether a restock is needed or not.

RUN EXPIRY DATE AUDITOR

Disaster Relief System

[DASHBOARD](#)
[LOGOUT](#)

Expiry Date Auditor

Monitors relief shelf-lives and flags resources needing priority queue dispatching.

ITEM ID	ITEM NAME	CATEGORY	QUANTITY	RESTOCK STATUS	EXPIRATION DATE	STATUS / DAYS LEFT	UPDATE STOCK	ACTIONS
1	brown rice	grains(kg)	0	!!RESTOCK NEEDED!!	2027-12-05	460 DAYS REMAINING	Amount + Add - Remove	Edit Delete
2	cookies	dry	23	STOCK AVAILABLE	2026-09-10	URGENT (9 DAYS LEFT)	Amount + Add - Remove	Edit Delete
3	apples	fruit	5	!!RESTOCK NEEDED!!	2026-09-02	URGENT (1 DAYS LEFT)	Amount + Add - Remove	Edit Delete

There's also an option to update, add, or remove stock quantities. Eg. Previously we had this

7	chips	dry	15	STOCK AVAILABLE	2026-09-07	URGENT (6 DAYS LEFT)	Amount	+ Add	- Remove	Edit	Delete
---	-------	-----	----	-----------------	------------	----------------------	--------	-------	----------	------	--------

If we want to remove 5 of this item, then

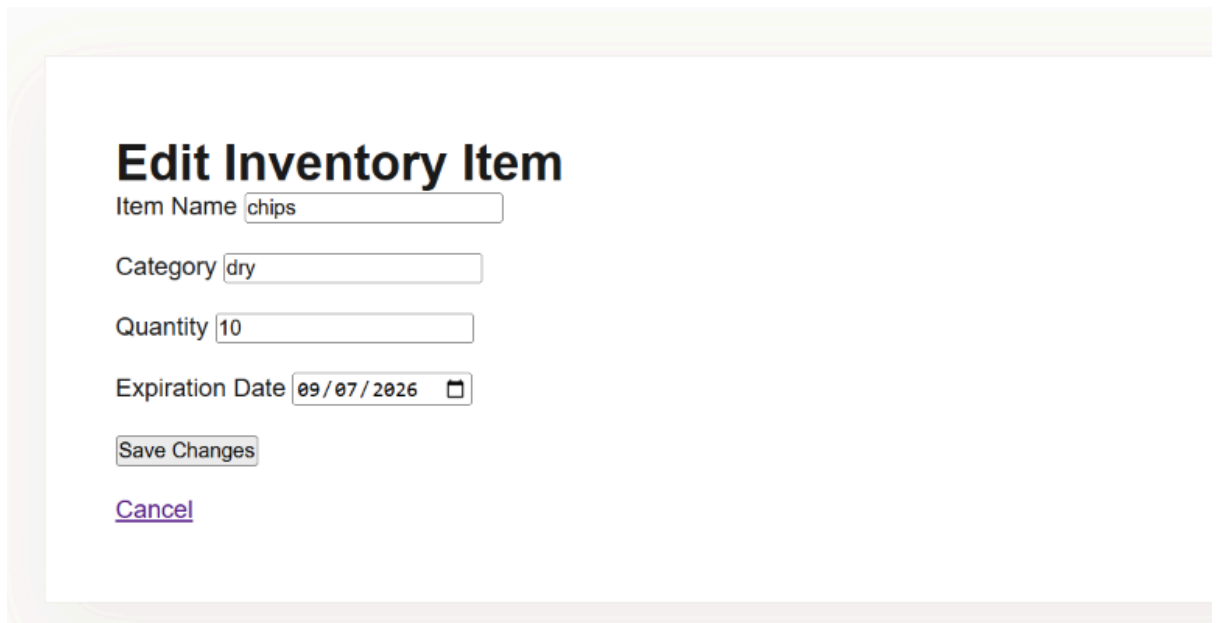
7	chips	dry	15	STOCK AVAILABLE	2026-09-07	URGENT (6 DAYS LEFT)	5	+ Add	- Remove	Edit	Delete
---	-------	-----	----	-----------------	------------	----------------------	--------------------------------	-------	----------	------	--------

After updating, we can see

7	chips	dry	10	!!RESTOCK NEEDED!!	2026-09-07	URGENT (6 DAYS LEFT)	Amount	+ Add	- Remove	Edit	Delete
---	-------	-----	----	--------------------	------------	----------------------	--------	-------	----------	------	--------

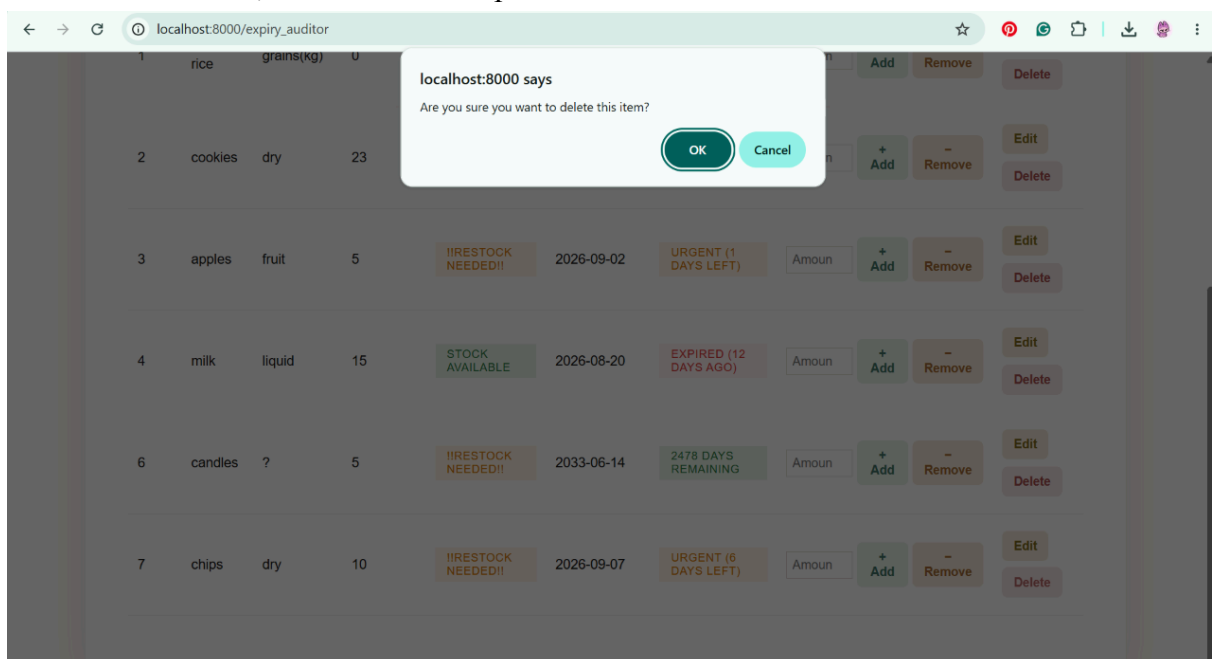
Additionally, we can edit or delete inventory items using the edit and delete buttons at the rightmost corner

Edit inventory page:



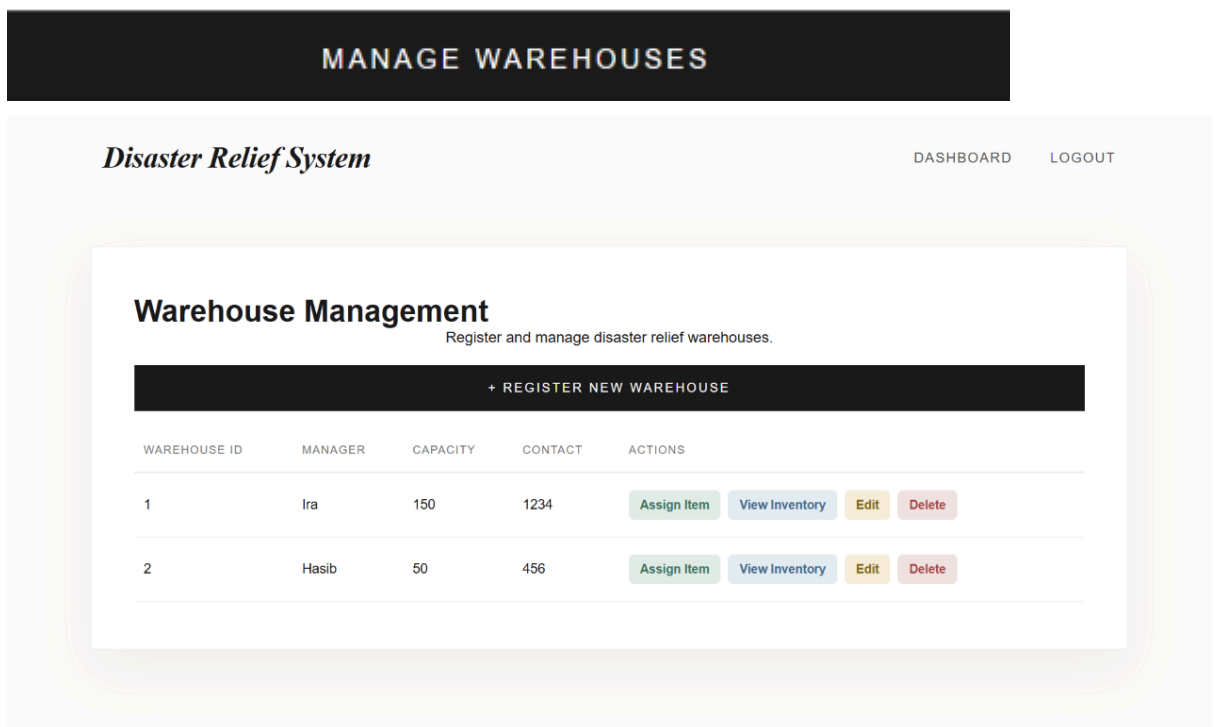
The screenshot shows a web form titled "Edit Inventory Item". It contains several input fields: "Item Name" with the value "chips", "Category" with the value "dry", "Quantity" with the value "10", and "Expiration Date" with the value "09/07/2026" and a calendar icon. Below the fields are two buttons: "Save Changes" and a purple "Cancel" link.

If we want to delete, then this shows up

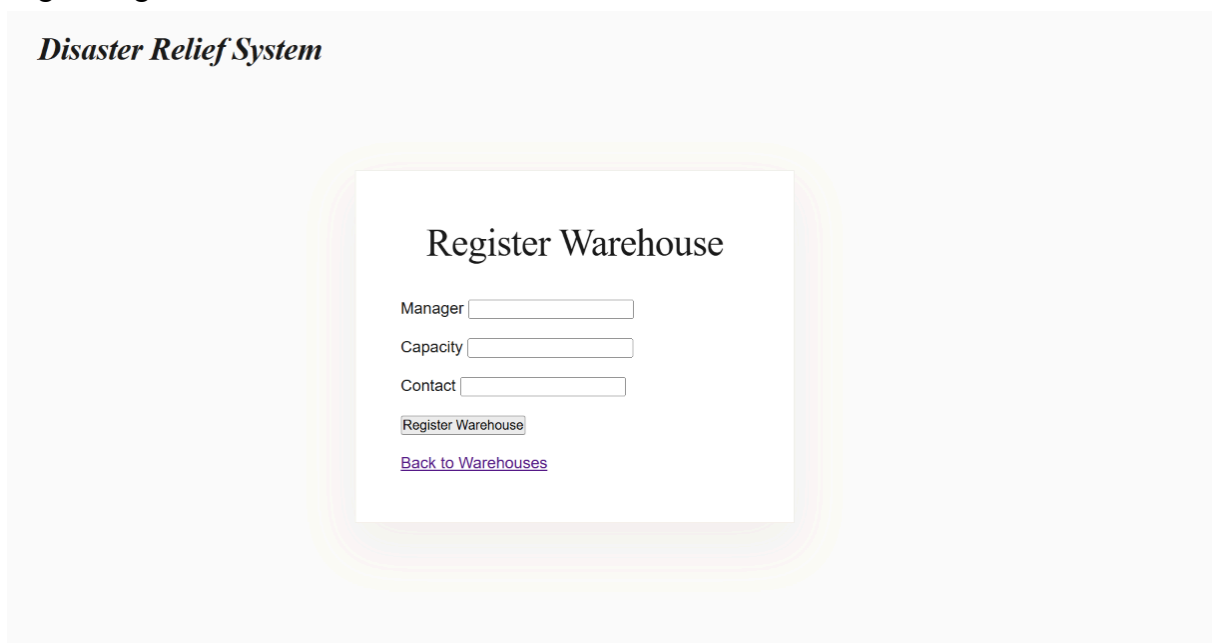


3. Manage warehouse page

This page allows us to register warehouses, view the registered warehouses, assign inventory items to a warehouse, view a warehouse's inventory stock, edit and delete a warehouse.



Registering a warehouse:



If we want to add a new warehouse, we have to register it first. Then it will show up in the warehouse management panel.

After that, we can assign items to that warehouse by selecting available inventory items and designating them to a shelf location.

Disaster Relief System

WAREHOUSES LOGOUT

Assign Item to Warehouse 1

Inventory Item

Shelf Location

[Cancel](#)

We can also view the inventory of a warehouse:

Disaster Relief System

DASHBOARD LOGOUT

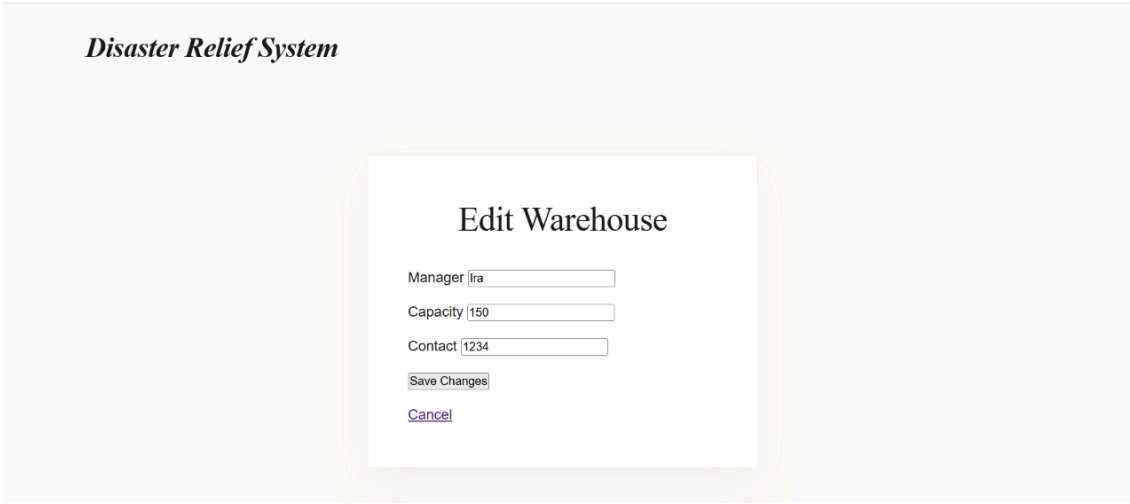
Inventory in

Warehouse ID: 1

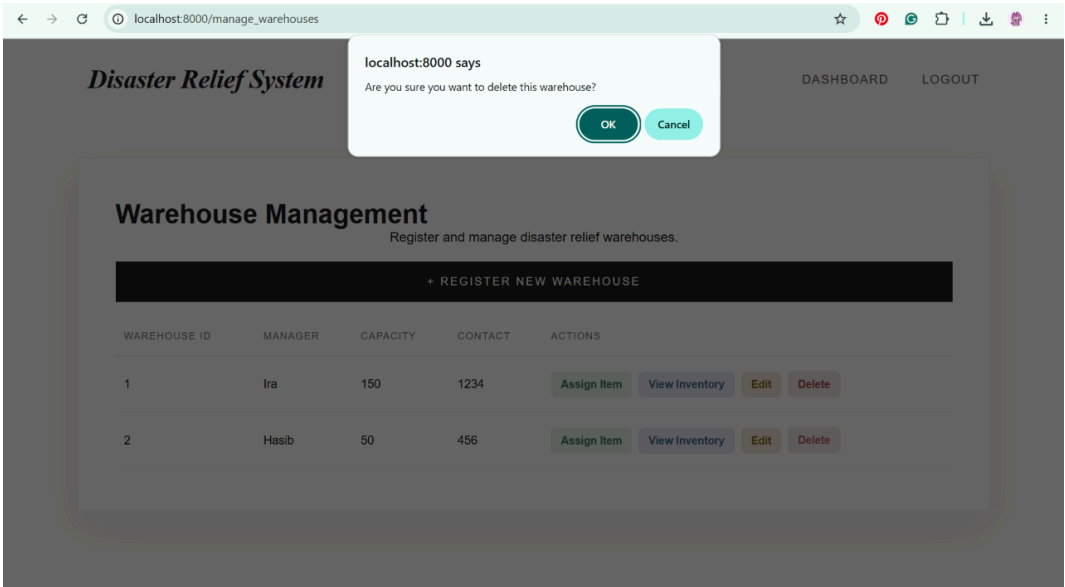
ITEM ID	ITEM NAME	CATEGORY	QUANTITY	EXPIRATION DATE	SHELF LOCATION
3	apples	fruit	5	2026-09-02	A1
6	candles	?	5	2033-06-14	C3

[← BACK TO WAREHOUSES](#)

Editing a warehouse:



Deleting a warehouse:



Backend Development

Briefly discuss about Backend Development and add relevant Screenshots (if required) by mentioning Individual Contributions

Contribution of ID : 24301374, Name : Shadeed Aarshan

We used Python with Flask for our project. I developed the login route for customers to enter their username and password. Following is the query running in the login route

```
sql = "SELECT * FROM User WHERE username = %s AND password = %s AND isActive = 1"
cursor.execute(sql, (username, password))
user = cursor.fetchone()

if user:
    cursor.execute("SELECT * FROM Admin WHERE Username = %s", (username,))
    is_admin = cursor.fetchone()

    cursor.execute("SELECT * FROM Customer WHERE Username = %s", (username,))
    is_customer = cursor.fetchone()

    session['username'] = username
    if is_admin:
        session['role'] = 'Admin'
    elif is_customer:
        cursor.execute("SELECT * FROM Volunteers WHERE Username = %s", (username,))
        if cursor.fetchone():
            session['role'] = 'Volunteer'
        else:
            session['role'] = 'Customer'
    else:
        session['role'] = 'User'
```

I developed the feature for customers to be able to apply as volunteers. This updates the volunteer table. Following is the query in the /become_volunteer route

```

        try:
            sql = """
                INSERT INTO Volunteers
                (Username, FullName, AvailabilityStatus,
                VolunteerType, specialty, certificationLevel,
                medicalSpecialty, license)
                VALUES (%s, %s, 'Pending', %s,
                %s, %s, %s, %s)
            """

            cursor.execute(sql,
            (session['username'], full_name, vol_type,
            specialty, cert_level, med_specialty,
            license_num))
            conn.commit()

            return

```

Admin can monitor volunteers through the /monitor_volunteer route

```

        sql = """
            SELECT v.VolunteerID, v.Username,
            v.FullName, v.specialty, v.AvailabilityStatus,
            IFNULL(history_counts.hrs, 0) AS
            total_hours
            FROM Volunteers v
            LEFT JOIN (
                SELECT VolunteerID,
                SUM(hours_earned) as hrs
                FROM deployment_history GROUP BY
                VolunteerID
            ) history_counts ON v.VolunteerID =
            history_counts.VolunteerID
            ORDER BY field(v.AvailabilityStatus,
            'Pending') DESC, total_hours DESC
        """

```

Connecting with the previous feature, I developed the admin feature of approving/removing volunteers. Following pictures are the queries ran by the /approve_volunteer and /remove_volunteer routes respectively.

/approve_volunteer

```
try:
    cursor.execute("UPDATE Volunteers SET
AvailabilityStatus = 'Available' WHERE
VolunteerID = %s", (volunteer_id,))
    conn.commit()
```

/remove_volunteer

```
try:
    cursor.execute("DELETE FROM
volunteers_deployedto_dzones WHERE VolunteerID =
%s", (volunteer_id,))
    cursor.execute("DELETE FROM Volunteers
WHERE VolunteerID = %s", (volunteer_id,))
    conn.commit()
```

I developed the admin feature of being able to deploy/release volunteers to and from disaster zones. Admin can also view current deployments.

/deploy_volunteers

```

        cursor.execute("SELECT VolunteerID,
FullName, specialty FROM Volunteers WHERE
AvailabilityStatus = 'Available'")
        available_vols = cursor.fetchall()

        cursor.execute("SELECT ZoneId, name,
location, severity FROM disasterzones")
        zones = cursor.fetchall()

        user_specialty = ""
        if selected_volunteer_id:
            cursor.execute("SELECT specialty FROM
Volunteers WHERE VolunteerID = %s",
(selected_volunteer_id,))
            result = cursor.fetchone()
            if result:
                user_specialty = result['specialty']

```

/process_deployment

```

try:
    sql_deploy = """
        INSERT INTO
volunteers_deployedto_dzones (`VolunteerID`,
`ZoneId`, `current_role`, `hours_contributed`,
`deployed_at`)
        VALUES (%s, %s, %s, 0, NOW())
    """
    cursor.execute(sql_deploy,
(volunteer_id, zone_id, role))
    cursor.execute("UPDATE Volunteers SET
AvailabilityStatus = 'Deployed' WHERE
VolunteerID = %s", (volunteer_id,))
    conn.commit()
    return

```

/active_deployments

```

    sql = """
        SELECT d.VolunteerID, d.ZoneId,
d.current_role, d.hours_contributed,
d.deployed_at,
            v.FullName, dz.name AS zone_name,
dz.location
        FROM volunteers_deployedto_dzones d
        JOIN Volunteers v ON d.VolunteerID =
v.VolunteerID
        JOIN disasterzones dz ON d.ZoneId =
dz.ZoneId
    """
    cursor.execute(sql)
    deployments = cursor.fetchall()

```

/release_volunteer


```

try:
    cursor.execute("SELECT deployed_at FROM
volunteers_deployedto_dzones WHERE VolunteerID =
%s AND ZoneId = %s", (volunteer_id, zone_id))
    deployment = cursor.fetchone()

    if deployment:
        cursor.execute("SELECT
TIMESTAMPDIFF(HOUR, %s, NOW()) AS hours_spent",
(deployment['deployed_at'],))
        result = cursor.fetchone()
        calculated_hours =
result['hours_spent'] if result and
result['hours_spent'] is not None else 0
        if calculated_hours < 1:
            calculated_hours = 1

        cursor.execute("""
            INSERT INTO deployment_history
(VolunteerID, hours_earned)
            VALUES (%s, %s)
        """, (volunteer_id,
calculated_hours))

        cursor.execute("DELETE FROM
volunteers_deployedto_dzones WHERE VolunteerID =
%s AND ZoneId = %s", (volunteer_id, zone_id))
        cursor.execute("UPDATE Volunteers
SET AvailabilityStatus = 'Available' WHERE
VolunteerID = %s", (volunteer_id,))

    conn.commit()
    return

```

Finally, I developed the customer feature to be able to toggle their availability. If they are unavailable, they cannot be deployed.

/volunteer/toggle_availability

```

try:
    cursor.execute("UPDATE Volunteers SET
AvailabilityStatus = %s WHERE Username = %s AND
AvailabilityStatus != 'Deployed'", (new_status,
username))
    conn.commit()

```

Contribution of ID : 24301348, Name : Kazi Aryan Rahman

I developed the Disaster Zone & Shipment Tracking feature, which lets admins declare new disaster zones and assign them a severity level and a source warehouse. Zones are listed ordered by severity so the most critical ones surface first, and admins can update a zone's operational status as relief progresses.

/manage_zones

```

#based on severity highest one is shown
cursor.execute("SELECT * FROM disasterzones ORDER BY severity DESC")
zones = cursor.fetchall()

```

/add_zone

```

cursor.execute("SELECT MAX(ZoneId) FROM disasterzones")
max_id = cursor.fetchone()[0]
new_id = 1 if max_id is None else max_id + 1

sql = """
    INSERT INTO disasterzones (ZoneId, name, location, severity, warehouseID, status)
    VALUES (%s, %s, %s, %s, %s, %s)
    """

cursor.execute(sql, (new_id, name, location, severity, warehouse_id, status))
conn.commit()

```

/update_zone/<zone_id>

```

cursor.execute("UPDATE disasterzones SET status = %s WHERE ZoneId = %s", (new_status, zone_id))
conn.commit()

```

Building on that, I developed the Relief Shipment Dispatcher. It pulls the items currently available in a zone's assigned warehouse, then on submission checks that enough stock exists, deducts the dispatched quantity from inventory, records the dispatch in a permanent shipment log, and automatically updates the zone's status to 'Dispatched' with a timestamp.

/dispatch_shipment/<zone_id> — GET (load items available in the zone's warehouse)

```
fetch_inventory_sql = """
    SELECT i.ItemId, i.ItemName, i.Quantity, i.Category
    FROM inventoryitems i
    JOIN warehouse_contains_inventoryitems wci ON i.ItemId = wci.ItemId
    JOIN warehouses w ON w.WID = wci.WID
    JOIN disasterzones dz ON dz.warehouseID = w.WID
    WHERE dz.ZoneId = %s AND i.Quantity > 0
    """
cursor.execute(fetch_inventory_sql, (zone_id,))
available_items = cursor.fetchall()
```

/dispatch_shipment/<zone_id> — POST (validate stock, deduct inventory, log shipment, update zone status)

```
# check if enough stock exists first
cursor.execute("SELECT Quantity FROM inventoryitems WHERE ItemId = %s", (item_id,))
item_data = cursor.fetchone()

if not item_data or item_data['Quantity'] < dispatch_qty:
    return "Error: Insufficient stock for this dispatch."

# get the assigned warehouse ID for this zone
cursor.execute("SELECT warehouseID FROM disasterzones WHERE ZoneId = %s", (zone_id,))
zone_data = cursor.fetchone()
warehouse_id = zone_data['warehouseID'] if zone_data else 0

# deduct from inventory
update_sql = "UPDATE inventoryitems SET Quantity = Quantity - %s WHERE ItemId = %s"
cursor.execute(update_sql, (dispatch_qty, item_id))

# log the shipment dispatch into our new table
log_sql = """
    INSERT INTO shipmentlog (ItemId, ZoneId, WID, QuantityShipped, DispatchedBy, DispatchedAt)
    VALUES (%s, %s, %s, %s, %s, NOW())
    """
cursor.execute(log_sql, (item_id, zone_id, warehouse_id, dispatch_qty, session['username']))

# automatically update zone status to dispatched
cursor.execute(
    "UPDATE disasterzones SET status = 'Dispatched', dispatchTimeStamp = NOW() WHERE ZoneId = %s",
    (zone_id,)
)

conn.commit()
```

Finally, I developed the Live Disaster Zone Analytics & Resource Report — a single aggregation query joining zones with their deployed volunteers and shipment log to report, per zone, active personnel, live accumulated volunteer hours, and total supplies dispatched, ordered by severity.

/zone_analytics

```
analytics_sql = """
SELECT
    dz.ZoneId,
    dz.name,
    dz.location,
    dz.severity,
    dz.status,
    COUNT(DISTINCT vd.VolunteerID) AS active_personnel,
    COALESCE(SUM(TIMESTAMPDIFF(HOUR, vd.deployed_at, NOW())), 0) AS live_hours_contributed,
    COALESCE(sl.total_shipped, 0) AS total_items_dispatched
FROM disasterzones dz
LEFT JOIN volunteers_deployedto_dzones vd ON dz.ZoneId = vd.ZoneId
LEFT JOIN (
    SELECT ZoneId, SUM(QuantityShipped) AS total_shipped
    FROM shipmentlog GROUP BY ZoneId
) sl ON dz.ZoneId = sl.ZoneId
GROUP BY dz.ZoneId, dz.name, dz.location, dz.severity, dz.status, sl.total_shipped
ORDER BY dz.severity DESC, active_personnel DESC
"""
cursor.execute(analytics_sql)
analytics = cursor.fetchall()
```

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I developed the features that handled the viewing, adding, editing, and deleting of inventory items. To add an item to the inventory:

```
try:
    sql = """
        INSERT INTO InventoryItems
        (ItemName, Quantity, Category, ExpirationDate)
        VALUES (%s, %s, %s, %s)
    """

    cursor.execute(
        sql,
        (item_name, quantity, category, expiration_date)
    )
    conn.commit()
```

To edit an item:

```
sql = """
    UPDATE InventoryItems
    SET ItemName = %s, Category = %s, Quantity = %s,
    ExpirationDate = %s
    WHERE ItemId = %s
    """
cursor.execute(sql,(item_name,category,quantity,expiration_date,item_id))
conn.commit()
cursor.close()
conn.close()

return redirect(url_for('expiry_auditor'))
cursor.execute(
    "SELECT * FROM InventoryItems where ItemId =%s",
    (item_id,)
)
item = cursor.fetchone()
```

To delete an item:

```
cursor.execute(
    "DELETE FROM InventoryItems where ItemId= %s",
    (item_id,)
)
```

The added inventory items can be seen in the expiry date auditor, which runs the following query:

```
sql = """
    SELECT ItemId, ItemName, ExpirationDate, Quantity, Category,
    DATEDIFF(ExpirationDate, CURDATE()) AS days_remaining,
    CASE
        WHEN Quantity<=10 THEN '!!RESTOCK NEEDED!!'
        ELSE 'Stock Available :)'
    END AS RestockStatus
    FROM InventoryItems
    ORDER BY ItemId ASC,ExpirationDate ASC
    """
cursor.execute(sql)
items = cursor.fetchall()
```

From the expiry date auditor, we can update our inventory stock:

```

cursor.execute(
    "SELECT Quantity FROM InventoryItems WHERE ItemId = %s",
    (item_id,)
)
item = cursor.fetchone()

if item:
    current_quantity = item['Quantity']

    if action== 'add':
        new_quantity = current_quantity + amount
    elif action== 'remove':
        new_quantity = current_quantity - amount

        if new_quantity<0 :
            new_quantity = 0
    else:
        new_quantity = current_quantity
    cursor.execute(
        "UPDATE InventoryItems SET Quantity =%s WHERE ItemId = %s",
        (new_quantity,item_id)
    )
conn.commit()

```

Moreover, one of my features had to manage warehouses and their corresponding inventory stock by shelf number. To view warehouses:

```

cursor.execute(
    "SELECT *FROM Warehouses order by WID ASC"
)

warehouses = cursor.fetchall()

```

To add/register a warehouse:

```

cursor.execute(
    """INSERT INTO Warehouses (Manager,Capacity,Contact)
    values (%s,%s,%s)
    """, (manager,capacity,contact)
)

conn.commit()

```

To edit warehouses:

```
cursor.execute(
    """
    UPDATE Warehouses
    set Manager=%s, Capacity= %s, Contact= %s
    where WID = %s
    """, (manager, capacity, contact, wid)
)
conn.commit()
```

To delete a warehouse:

```
cursor.execute(
    "DELETE from Warehouses where WID=%s",
    (wid,)
)
```

To assign an inventory item to a warehouse:

```
cursor.execute(
    """
    INSERT INTO warehouse_contains_inventoryitems (ItemId,WID, shelf_location)
    values (%s,%s,%s)
    """,
    (item_id, wid, shelf_location)
)
```

The query below allows us to choose the inventory items from a (dropdown) list:

```
cursor.execute(
    """
    SELECT ItemId, ItemName
    from InventoryItems
    ORDER by ItemName ASC
    """
)
items = cursor.fetchall()
```

Finally, to view the inventory in a warehouse:

```
cursor.execute(
    """
    SELECT i.ItemId, i.ItemName, i.Category, i.Quantity,
    i.ExpirationDate, wci.shelf_location
    FROM warehouse_contains_inventoryitems wci
    JOIN inventoryitems i ON wci.ItemId = i.ItemId
    WHERE wci.WID = %s
    """, (wid,)
)
items = cursor.fetchall()
```

Source Code Repository

Google Drive link with Project Source Code:

Google Drive:

https://drive.google.com/drive/folders/1jyuPn6trXx8PWMf4x6A1wl15ViQ7elGS?usp=drive_link

GitHub:

https://github.com/oishee296/CSE370_Group08_S11_Summer2026

Conclusion

Through this project, we learned how to handle a real-life problem using an RDBMS to create relationships between different entities and using SQL queries to store, retrieve and manage data. We also learned to connect our database with a Flask application and create features like registration, login, deployment, and updating. To sum up, our project helped us understand how to deal with the frontend, backend, and database all together to build a working system.

References